Swimming pool heating using data-center waste heat

Engineering calculation and numerical simulations, 500 m³

Reference scenarios only: not a calibrated or commissioned system design.

Illustrative geometry is 25 × 12.5 × 1.6 m, water area 312.5 m². The model includes evaporation, convection, radiation, ground/structure losses, make-up water and auxiliary load. Building heating, ventilation and dehumidification energy are outside this balance.

Governing equations and substitution

$C = \rho V c_p = 997 \times 500 \times 4182 = 2\,084\,727\,000$ J/K = 0,57909 MWh/K.

$E = C (28 - 16) = 6.9491$ MWh.

$C \cdot dT/dt = Q_{dod} - Q_{vypar} - Q_{konv} - Q_{rad} - Q_{pod} - Q_{dop} - Q_{ostatné}$.

$p_{sat}(T) = 0{,}61094 \exp[17{,}625\, T/(T+243{,}04)]$ kPa; $p_{air} = RH\, p_{sat}(T_{air})$.

$m_{evap} = 4e\text{-}5\, A \max(p_{sat}(T)-p_{air}, 0)\, Fa\, (1-0{,}9 \cdot cover)$ kg/s.

$Q_{evap} = m_{evap} h_{fg}$; $h_{fg} = 2{,}44$ MJ/kg.

$Q_{conv} = h\, A\, (T-T_{air})\, (1-0{,}75 \cdot cover)$, $h = 3$ W/(m² K).

$Q_{rad} = \varepsilon \sigma A ((T+273{,}15)^4-(T_{wall}+273{,}15)^4)\, (1-0{,}75 \cdot cover)$.

$Q_{ground} = UA\, (T-T_{ground})$; $Q_{makeup} = m_{evap} c_p\, (T-T_{makeup})$.

$Q_{out,HP} = Q_{source,DC} + W_{electric}$; $COP = Q_{out,HP} / W_{electric}$.

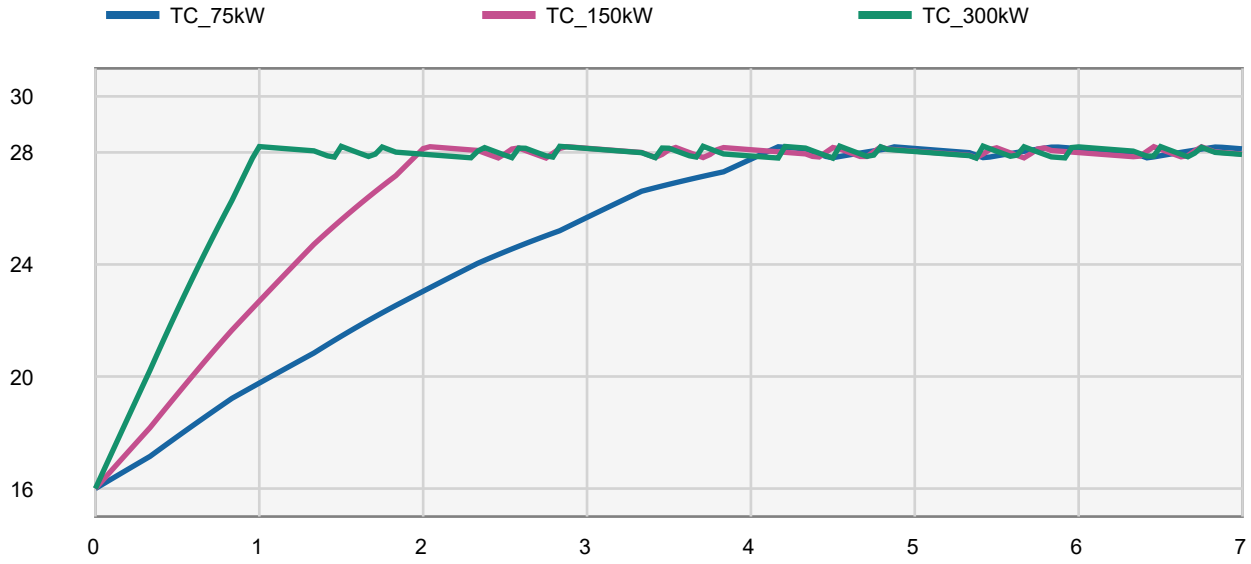
Thermostat switches on at 27.8 °C and off at 28.2 °C. Open from 08:00–20:00; night cover thereafter. Cover evaporation reduction 90%, ground conductance, COP 4 and relative humidity are scenario assumptions.

Rated IT capacity 10 MW, utilization 82%, recoverable 85% of IT load implies 6.97 MW potential, not pool-delivered heat. The source branch to the pool is limited to 500 kW. Direct heat exchange assumes 35 °C coolant with 3 K approach; the heat-pump case represents a cooler 25 °C source.

Simulation results

Scenario	28 °C [h]	Heat [MWh]	HP electricity [MWh]
TC_75kW	98.18	9.67	2.42
TC_150kW	47.43	10.86	2.72
TC_300kW	23.41	11.53	2.88
VYMENNIK_150kW	47.43	10.86	0.00
TC_150kW_bez_krytu	46.86	12.01	3.00
TC_150kW_RH40	50.84	12.90	3.22
TC_150kW_RH80	45.40	9.14	2.28

Pool-water temperature



Heat-flow components at 28 °C water

	08–20 [kW]	20–08 [kW]
Evaporation	41.91	2.10
Convection	-0.94	-0.23
Radiation	1.83	0.46
Structure	6.40	6.40
Make-up	0.93	0.05
Other	2.00	2.00
Total	52.13	10.77

Checks and limitations

The study lacks measured coolant temperatures/flows, equipment COP map, actual natatorium psychrometrics, heat-exchanger transfer limits, and hydraulics. Independent IT fail-safe cooling and separated pool-water circuits are essential.

Numerical integration: forward Euler, 300 s step, 7 days. 60-s independent step-size comparison.

Pool balance residual = 7.150e-14 MWh.

Heat pump first-law residual = -1.914e-13 MWh.

Time-grid difference, final pool temperature = 0.21412 °C.

First 28 °C threshold crossing: 60-s versus 300-s step difference = 0.61 minutes.

The final sample can differ with relay-cycle phase; the first threshold crossing is a more meaningful grid check.

Holding load: open 52.13 kW, covered 10.77 kW, average 31.45 kW.

References / Zdroje / Источники

ASHRAE Handbook 2023, HVAC Applications, Ch. 6 Indoor Swimming Pools:

https://handbook.ashrae.org/Handbooks/A23/SI/A23_Ch06/a23_ch06_si.aspx

EnergyPlus Engineering Reference, Indoor Swimming Pool:

https://energyplus.net/assets/nrel_custom/pdfs/pdfs_v22.1.0/EngineeringReference.pdf

Project data: docs/Assumptions_and_Parameters.md and scenario JSON/CSV in results/pool_500m3_v01.

Extended mathematical annex: boundaries, derivations and validation

1. Input assumptions and dimensional analysis

The indoor pool is treated as one perfectly mixed thermal mass. Its 25 × 12.5 × 1.6 m geometry is an assumed example, not a standard volume for every pool. Water heats from 16 to 28 °C. Room air is 29 °C at 60% relative humidity, walls 27 °C, ground 20 °C, and make-up water 15 °C.

$V = 25 \times 12.5 \times 1.6 = 500\ m^3$, $A = 25 \times 12.5 = 312.5\ m^2$.

$m = \rho V = 997\ kg/m^3 \times 500\ m^3 = 498500\ kg$.

$C = m\ cp = 498500\ kg \times 4182\ J/(kg\ K) = 2{,}084{,}727{,}000\ J/K$.

$E = C\ (28 - 16)\ K = 25{,}016{,}724{,}000\ J = 6.94909\ MWh$.

$1\ MWh = 3.6e9\ J$; $C = 0.57909\ MWh/K$.

$t_{min,ideal} = E / P = 92.65\ h$ at 75 kW; 46.33 h at 150 kW; 23.16 h at 300 kW.

Time-dependent heat flows into and out of the pool must be added to stored-water energy. $E = C\ deltaT$ is only the heat-storage requirement, not the total duty including thermal losses. It also excludes heating the natatorium building.

First law: $C\ dT/dt = Qheat(t) - Qevap(t) - Qconv(t) - Qrad(t) - Qground(t) - Qmakeup(t) - Qother(t)$.

Every Q is W (J/s), so $dT/dt = (J/s)/(J/K) = K/s$.

2. Derivation of operating losses and sensitivity

The simplified ASHRAE evaporation equation is empirical and expects partial pressures in kPa. Activity factor 1.0 is representative of an active public pool; 0.5 is used for an unoccupied pool. The assumed cover reduces evaporation by 90% and convection/radiation by 75%. Measured or manufacturer performance must replace these hypothetical values.

psat(T) = 0.61094 exp(17.625 T/(T+243.04)) kPa.

pa = RH psat(Tair); RH = 0.60; Tair = 29 °C.

mevap = 4e-5 A max(psat(T) - pa, 0) Fa (1 - 0.90 cover) kg/s.

Qevap = mevap × 2.44e6 J/kg.

Qconv = 3 A (T - 29) (1 - 0.75 cover) W.

Qrad = 0.95 σ A [(T + 273.15)^4 - (27 + 273.15)^4] (1 - 0.75 cover) W.

Qground = 800 (T - 20) W; Qmakeup = mevap 4182 (T - 15) W.

Qother = 2000 W; σ = 5.670374419e-8 W/(m² K⁴).

Keep signed heat exchange in the energy balance. With 28 °C pool water and 29 °C room air, convection is negative, so the warmer air heats the pool. The nonnegative clamp is appropriate only for the potential evaporation rate.

At 28 °C and 60% RH: Qloss,open = 52.132 kW, Qloss,covered = 10.765 kW.

12 h open + 12 h covered, 28 °C throughout: Ehold,day ≈ 0.755 MWh/day.

Equivalent mean holding power: 31.449 kW.

3. Heat pump, direct heat exchange and the data-center boundary

The IT nameplate is not the pool heating capacity: 10 MW × 0.82 × 0.85 = 6.97 MW of potential recoverable IT heat, while the dedicated source-side branch is only 500 kW. The assumed PUE of 1.20 is not multiplied into IT waste heat again: doing so would risk double counting infrastructure heat.

$Q_{DC,usable} = P_{IT,rated} \times utilization \times recoverable_fraction = 6.97\ MW$.

Direct HX: $Q_{pool} \le \min(Q_{available}, Q_{HX,nameplate})$; verify coolant source temperature and 3 K approach.

Heat pump: $Q_{out} = Q_{source} + W_{electric}$; $COP = Q_{out}/W_{electric}$.

$W_{electric} = Q_{out}/COP$; $Q_{source} = Q_{out}(1 - 1/COP)$.

At $Q_{out}=150$ kW and COP=4: $W_{electric}=37.5$ kW; $Q_{source}=112.5$ kW.

At COP=3/4/5 and fixed $Q_{out}=150$ kW, compressor electricity=50/37.5/30 kW.

For 7-day baseline $Q_{out}=10.8625$ MWh:

$W(COP=3)\approx3.6208$ MWh; $W(COP=4)=2.7156$ MWh; $W(COP=5)\approx2.1725$ MWh.

At COP=4: source waste heat = 8.1469 MWh; compressor electricity = 2.7156 MWh.

COP=4 is an assumed constant, not a part-load or temperature-dependent equipment map. Direct heat exchange avoids compressor electricity, but does not make circulation pumps, valves and control consume zero power.

Simple electricity-only energy cost formula: $OPEX_{el} = E_{out} \times electricity_price / COP$.

Avoid comparing direct HX and HP without circulation, capacity, seasonality, CAPEX and opportunity costs.

4. Integration, executed results and acceptance checks

The Python model uses explicit Euler with a 300-s time step and a 60-s sensitivity run. Simulink uses a continuous integrator with the ode4 solver and fixed 60-s step. Hysteresis changes the phase of on/off cycles and can shift the end-point temperature. Validate energy balance, first target crossing, source capacity, and operating limits rather than relying on a single final sample.

Euler: T[k+1] = T[k] + dt/C (Qin[k] - Qloss[k]); dt=300 s.

Check: C (Tfinal - T0) - Σ(Qin-Qloss) dt ≈ 0.

Python first 28 °C at 150 kW = 47.434 h.

Python dt=60 s vs dt=300 s: first-crossing difference = 0.607 min.

Executed Simulink solver: fixed-step ode4, dt=60 s, thermostat ON at 27.8 °C, OFF at 28.2 °C.

Simulink case	First 28 °C [h]	Relative residual
TC_75kW	97.883	6.57728e-05
TC_150kW	47.317	0.000117025
TC_300kW	23.350	0.000217481
TC_150kW_bez_plachty	46.833	7.99335e-09

Next work requires measured coolant temperatures and flow, an exchanger pressure-drop/thermal design, pool-water separation, COP(T) performance map and annual operation. This extension is not yet wired into the existing v85 Stateflow dispatcher and does not modify its physical or safety contracts.

Sources: ASHRAE HVAC Applications 2023 ch. 6, Indoor Swimming Pools; EnergyPlus Engineering Reference.

Project artifacts: MATLAB/Simulink model, executed CSV, JSON, validation logs and three-language report, 2026-10-23.